2.26 Differentiation: Implicit Differentiation

Worked Solutions 1–30

1 a)

$2y\frac{dy}{dx} + 1.$

1 b)

$x\frac{dy}{dx} + y.$

1 c)

$x^2\frac{dy}{dx} + 2xy.$

1 d)

$e^x \cos y \frac{dy}{dx} + e^x \sin y.$

1 e)

$-\sin x\, e^y + e^y \frac{dy}{dx} \cos y.$

1 f)

$\frac{x}{y}\frac{dy}{dx} + \ln y.$

1 g)

$2^x\frac{dy}{dx} + (\ln 2) \times 2^x y.$

2 a)

$2y\frac{dy}{dx} = 8 \Rightarrow \frac{dy}{dx} = \frac{8}{2y} = \frac{4}{y}.$

2 b)

$x\frac{dy}{dx} + y = 12 \Rightarrow \frac{dy}{dx} = -\frac{y}{x}.$

2 c)

$$3x^2 + 2xy\frac{dy}{dx} + y^2 - 2y\frac{dy}{dx} = 0$$

$$\Rightarrow (2xy - 2y)\frac{dy}{dx} = -3x^2 - y^2 \Rightarrow \frac{dy}{dx} = \frac{-3x^2 - y^2}{2xy - 2y}$$

2 d)

$\frac{1}{x} + 2y\frac{dy}{dx} = 2 \Rightarrow 2y\frac{dy}{dx} = 2 - \frac{1}{x} \Rightarrow \frac{dy}{dx} = \frac{2-\frac{1}{x}}{2y} = \frac{2x-1}{2xy}.$

2 e)

$$\begin{aligned} 10x - 2x\frac{dy}{dx} - 2y + 10y\frac{dy}{dx} &= 0 \\ \Rightarrow (-2x + 10y)\frac{dy}{dx} &= 2y - 10x \\ \Rightarrow \frac{dy}{dx} &= \frac{2y - 10x}{-2x + 10y} = \frac{y - 5x}{5y - x} \end{aligned}$$

2 f)

$$\begin{aligned} \cos x + e^y + xe^y\frac{dy}{dx} &= 0 \\ \Rightarrow xe^y\frac{dy}{dx} = -\cos x - e^y \Rightarrow \frac{dy}{dx} &= \frac{-\cos x - e^y}{xe^y} \end{aligned}$$

2 g)

$-x\sin y\frac{dy}{dx} + \cos y = 1$ so $\frac{dy}{dx} = \frac{1-\cos y}{-x\sin y} = \frac{\cos y - 1}{x\sin y}$.

2 h)

$$\begin{aligned} \frac{x}{y}\frac{dy}{dx} + \ln y + \frac{y}{x} + \ln x\frac{dy}{dx} &= 0 \\ (\frac{x}{y} + \ln x)\frac{dy}{dx} &= -\ln y - \frac{y}{x} \\ \Rightarrow \frac{dy}{dx} &= -\frac{\ln y + \frac{y}{x}}{\frac{x}{y} + \ln x} = -\frac{y^2 + xy\ln y}{x^2 + xy\ln x} \end{aligned}$$

3)

$2x + 2y\frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -\frac{2x}{2y} = -\frac{x}{y}$. $\left.\frac{dy}{dx}\right|_{(4,3)} = -\frac{4}{3}$.

$$\begin{aligned} y - 3 = -\tfrac{4}{3}(x - 4) &\Rightarrow 3y - 9 = -4x + 16 \\ &\Rightarrow 3y + 4x - 25 = 0 \end{aligned}$$

4)

$y = \ln x$ and $e^y = x$. $e^y\frac{dy}{dx} = 1 \Rightarrow \frac{dy}{dx} = \frac{1}{e^y} = \frac{1}{x}$.

5)

$y = a^x$ and $\ln y = \ln a^x \Rightarrow \ln y = x\ln a$. $\frac{1}{y}\frac{dy}{dx} = \ln a \Rightarrow \frac{dy}{dx} = \ln a \times y = \ln a \times a^x$.

6)

$y = \arcsin x$ and $\sin y = x$. So $1 - \sin^2 y = 1 - x^2$. $\cos y\frac{dy}{dx} = 1 \Rightarrow \frac{dy}{dx} = \frac{1}{\cos y} = \frac{1}{\sqrt{1-x^2}}$.

7)

$y = \arccos x$ and $\cos y = x$. So $1 - \cos^2 y = 1 - x^2$. $-\sin y \frac{dy}{dx} = 1 \Rightarrow \frac{dy}{dx} = -\frac{1}{\sin y} = -\frac{1}{\sqrt{1-x^2}}$.

8)

$y = \arctan x$ and $\tan y = x$. So $\tan^2 y + 1 = x^2 + 1$. $\sec^2 y \frac{dy}{dx} = 1 \Rightarrow \frac{dy}{dx} = \frac{1}{\sec^2 y} = \frac{1}{1+x^2}$.

9 a)

$2\cos 2y \frac{dy}{dx} - 1 = 0 \Rightarrow 2\frac{dy}{dx} - 1 = 0 \Rightarrow \frac{dy}{dx} = \frac{1}{2}$.

9 b)

When $x = 2$, $4y = 2 \Rightarrow y = \frac{1}{2}$. $x^2 \frac{dy}{dx} + 2xy = 1 \Rightarrow 4\frac{dy}{dx} + 2 = 1 \Rightarrow \frac{dy}{dx} = -\frac{1}{4}$.

9 c)

When $y = \ln 3$, $3x - x^2 + 4 = 0 \Rightarrow x = -1$ or $x = 4$. $xe^y \frac{dy}{dx} + e^y - 2x = 0$. $-3\frac{dy}{dx} + 3 + 2 = 0 \Rightarrow \frac{dy}{dx} = \frac{5}{3}$.
Or $12\frac{dy}{dx} + 3 - 8 = 0 \Rightarrow \frac{dy}{dx} = \frac{5}{12}$.

10)

$$x^2 \cos y \frac{dy}{dx} + 2x \sin y = 0$$
$$\Rightarrow \frac{dy}{dx} = \frac{-2x \sin y}{x^2 \cos y} = \frac{-2 \sin y}{x \cos y} = -\frac{2 \tan y}{x}$$

But $\sin y = \frac{4}{x^2}$ so $\tan y = \frac{4}{\sqrt{x^4 - 16}} \Rightarrow \frac{dy}{dx} = -\frac{8}{x\sqrt{x^4-16}}$.

11)

$$x^2 \sec^2 y \frac{dy}{dx} + 2x \tan y = 0$$
$$\Rightarrow \frac{dy}{dx} = \frac{-2x \tan y}{x^2 \sec^2 y} = \frac{-2 \tan y}{x \sec^2 y}$$

But $\tan y = \frac{1}{x^2}$ so $\tan^2 y + 1 = \frac{1}{x^4} + 1$.

$$\Rightarrow \frac{dy}{dx} = \frac{-\frac{2}{x^2}}{x(\frac{1}{x^4} + 1)} = \frac{-2}{x^3(\frac{1}{x^4} + 1)} = \frac{-2}{\frac{1}{x} + x^3} = -\frac{2x}{1 + x^4}$$

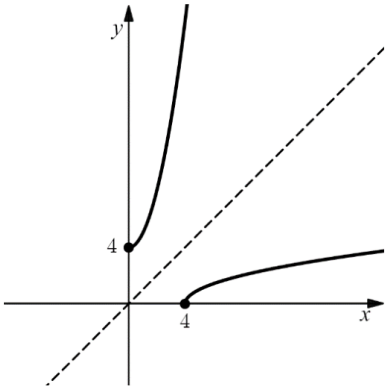
12 a)

$2x + 6y^2 \frac{dy}{dx} = 0$. Stat pt when $\frac{dy}{dx} = 0$ so $2x = 0 \Rightarrow x = 0$. So at $(0, 1)$.

12 b)

$\frac{dy}{dx} = -\frac{x}{3y^2}$ infinite gradient when $\frac{dx}{dy} = 0$. $-\frac{3y^2}{x} = 0$ so $y = 0$. So at $(-\sqrt{2}, 0)$ and $(\sqrt{2}, 0)$.

13 a)



13 b)

$f'(x) = 2x$ so $f'(3) = 6$. $x = y^2 + 4$ at $x = 13$, $y = 3$. $1 = 2y\frac{dy}{dx} \Rightarrow \frac{dy}{dx} = \frac{1}{2y} \Rightarrow \left.\frac{dy}{dx}\right|_{y=3} = \frac{1}{6}$. Therefore $f'(3) = \frac{1}{g'(13)}$ as req.

14)

$x = y(y-3) \Rightarrow x = y^2 - 3y$. $1 = 2y\frac{dy}{dx} - 3\frac{dy}{dx} \Rightarrow \frac{dy}{dx} = \frac{1}{2y-3} \Rightarrow \left.\frac{dy}{dx}\right|_{y=5} = \frac{1}{7}$.

15)

$2(x-2) = e^y\frac{dy}{dx}$. Tangent lines parallel to x-axis when $\frac{dy}{dx} = 0$. $2(x-2) = 0 \Rightarrow x = 2$ so $(2, \ln 2)$.

16)

$2xy\frac{dy}{dx} + y^2 + \frac{1}{2}y^{-\frac{1}{2}}\frac{dy}{dx} = 0$. Stat pt when $\frac{dy}{dx} = 0$. $y^2 = 0$ but $y > 0$ so no stat points.

17 a)

When $x = \frac{1}{4}$, $(\frac{1}{2} + y)^2 = \frac{1}{4} - y$. $\Rightarrow \frac{1}{4} + y + y^2 = \frac{1}{4} - y \Rightarrow y^2 + 2y = 0$. $\Rightarrow y(y+2) = 0$, therefore $y = 0$, $y = -2$. $(\frac{1}{4}, 0)$ and $(\frac{1}{4}, -2)$.

$$2(2x+y)(2 + \frac{dy}{dx}) = 1 - \frac{dy}{dx}$$

For $(\frac{1}{4}, 0)$: $2(\frac{1}{2})(2 + \frac{dy}{dx}) = 1 - \frac{dy}{dx}$.

$$\Rightarrow 2 + \frac{dy}{dx} = 1 - \frac{dy}{dx} \Rightarrow \frac{dy}{dx} = -\tfrac{1}{2}$$
$$y - 0 = -\tfrac{1}{2}(x - \tfrac{1}{4}) \Rightarrow 2y = -x + \tfrac{1}{4}$$
$$\Rightarrow 8y + 4x = 1$$

For $(\frac{1}{4}, -2)$: $2(\frac{1}{2} - 2)(2 + \frac{dy}{dx}) = 1 - \frac{dy}{dx}$.

$$\Rightarrow -6 - 3\frac{dy}{dx} = 1 - \frac{dy}{dx} \Rightarrow -7 = 2\frac{dy}{dx} \Rightarrow \frac{dy}{dx} = -\tfrac{7}{2}$$
$$y + 2 = -\tfrac{7}{2}(x - \tfrac{1}{4}) \Rightarrow 2y + 4 = -7x + \tfrac{7}{4}$$
$$\Rightarrow 8y + 16 = -28x + 7 \Rightarrow 8y + 28x = -9$$

17 b)

Solving $8y + 4x = 1$ and $8y + 28x = -9$ simultaneously gives $(-\frac{5}{12}, \frac{1}{3})$.

17 c)

When $y = 0$: $8y + 4x = 1 \Rightarrow x = \frac{1}{4}$. $8y + 28x = -9 \Rightarrow x = -\frac{9}{28}$. $\frac{1}{2} \times (\frac{1}{4} + \frac{9}{28}) \times \frac{1}{3} = \frac{2}{21}$ units^2.

18)

$$\begin{aligned} 10x + 12x\frac{dy}{dx} + 12y - 8y\frac{dy}{dx} &= 0 \\ \Rightarrow (12x - 8y)\frac{dy}{dx} &= -10x - 12y \\ \Rightarrow \frac{dy}{dx} &= \frac{-10x - 12y}{12x - 8y} = \frac{-5x - 6y}{6x - 4y} \end{aligned}$$

So $6x \neq 4y$, i.e. $y \neq \frac{3}{2}x$.

19 a)

LHS: $(1 + 1)^3 = 8$. RHS: $10 - 2 = 8$. So $(1, 1)$ lies on the curve.

19 b)

$3(x + y)^2(1 + \frac{dy}{dx}) = 20x - 2\frac{dy}{dx}$. At $(1, 1)$,

$$\begin{aligned} 3 \times 2^2(1 + \frac{dy}{dx}) &= 20 - 2\frac{dy}{dx} \\ \Rightarrow 12 + 12\frac{dy}{dx} &= 20 - 2\frac{dy}{dx} \\ \Rightarrow 14\frac{dy}{dx} &= 8 \Rightarrow \frac{dy}{dx} = \tfrac{4}{7} \\ y - 1 &= \tfrac{4}{7}(x - 1) \\ \Rightarrow 7y - 7 &= 4x - 4 \\ \Rightarrow 7y - 4x &= 3 \end{aligned}$$

20 a)

LHS: $1^3 + 2^2 = 5$. RHS: 5. Therefore the point $(1, 2)$ lies on the curve.

20 b)

$3x^2 + 2y\frac{dy}{dx} = 0$. At $(1, 2)$: $3 + 4\frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -\frac{3}{4}$. Normal has gradient $\frac{4}{3}$.

$$\begin{aligned} y - 2 = \tfrac{4}{3}(x - 1) &\Rightarrow 3y - 6 = 4x - 4 \\ &\Rightarrow 3y - 4x = 2 \end{aligned}$$

20 c)

Sub $y = \frac{4}{3}x + \frac{2}{3}$ into $x^3 + y^2 = 5$.

$$\begin{aligned} x^3 + (\tfrac{4}{3}x + \tfrac{2}{3})^2 &= 5 \\ \Rightarrow x^3 + \tfrac{16}{9}x^2 + \tfrac{16}{9}x + \tfrac{4}{9} &= 5 \end{aligned}$$

$$\Rightarrow x^3 + \tfrac{16}{9}x^2 + \tfrac{16}{9}x - \tfrac{41}{9} = 0$$
$$\Rightarrow 9x^3 + 16x^2 + 16x - 41 = 0$$
$$\Rightarrow (x-1)(9x^2 + 25x + 41) = 0$$

$b^2 - 4ac = 625 - 4 \times 9 \times 41 < 0$. So $x = 1$ is the only solution. Therefore the normal line does not intersect the equation again.

21 a)

LHS: $0^2 \times -1 + 0 = 0$. RHS: $\frac{1}{2} \times 0 = 0$ so $(0, \frac{3\pi}{2})$ lies on the curve.

21 b)

$$x^2 \cos y \frac{dy}{dx} + 2x \sin y - \sin y \frac{dy}{dx} = \tfrac{1}{2}$$
$$\Rightarrow (x^2 \cos y - \sin y)\frac{dy}{dx} = \tfrac{1}{2} - 2x \sin y$$
$$\Rightarrow \frac{dy}{dx} = \frac{\frac{1}{2} - 2x \sin y}{x^2 \cos y - \sin y} = \frac{1 - 4x \sin y}{2x^2 \cos y - 2 \sin y}$$
$$= \frac{4x \sin y - 1}{2 \sin y - 2x^2 \cos y} \text{ as req}$$

21 c)

$\left.\frac{dy}{dx}\right|_{(0, \frac{3\pi}{2})} = \frac{0-1}{-2-0} = \frac{1}{2}$. So $y - \frac{3\pi}{2} = \frac{1}{2}x \Rightarrow y = \frac{1}{2}x + \frac{3\pi}{2}$.

22 a)

LHS: $1^2 - 2\ln(1) = 1 - 0 = 1$: RHS. So $(1, 1)$ lies on the curve.

22 b)

$$2x - 2x\frac{1}{y}\frac{dy}{dx} - 2\ln y = 0 \Rightarrow x - \frac{x}{y}\frac{dy}{dx} - \ln y = 0$$
$$\Rightarrow \frac{x}{y}\frac{dy}{dx} = x - \ln y \Rightarrow \frac{dy}{dx} = \frac{y(x - \ln y)}{x} \text{ as req.}$$

22 c)

$\left.\frac{dy}{dx}\right|_{(1,1)} = \frac{1(1-\ln 1)}{1} = 1$. $y - 1 = -1(x-1) \Rightarrow y = -x + 2$.

23 a)

$4(x-5) - 2x\frac{dy}{dx} - 2y + 2(y+1)\frac{dy}{dx} = 0$. Stat pt when $\frac{dy}{dx} = 0$. $4(x-5) - 2y = 0 \Rightarrow y = 2x - 10$. Line meets the curve:

$$2(x-5)^2 - 2x(2x-10) + (2x-9)^2 = 1$$
$$2x^2 - 20x + 50 - 4x^2 + 20x + 4x^2 - 36x + 81 = 1$$
$$2x^2 - 36x + 130 = 0 \text{ gives } x = 5 \text{ or } x = 13$$

$y = 0$ or $y = 16$ so $(5, 0)$ and $(13, 16)$. P is $(13, 16)$.

23 b)

$\sqrt{8^2+16^2} = 8\sqrt{5}$.

24 a)

$(-\sqrt{2})^2 + \ln 1 = k$, therefore $k = 2$.

24 b)

$$2y\frac{dy}{dx} + \frac{1}{x} = 0 \Rightarrow 2y\frac{dy}{dx} = -\frac{1}{x}$$
$$\Rightarrow \frac{dy}{dx} = -\frac{1}{2xy}, \text{ therefore } x = 0 \text{ or } y = 0$$

But since $x > 0$ then $y = 0$. $0^2 + \ln x = 2 \Rightarrow x = e^2$. $(e^2, 0)$.

25 a)

LHS: $2 \times e^{4\ln 2} = 2 \times 2^4 = 32$: RHS so $(2, 4\ln 2)$ lies on the curve.

25 b)

$xe^y \frac{dy}{dx} + e^y = 0 \Rightarrow \frac{dy}{dx} = -\frac{e^y}{xe^y} = -\frac{1}{x}$. $\left.\frac{dy}{dx}\right|_{x=2} = -\frac{1}{2}$. Tangent line: $y - 4\ln 2 = -\frac{1}{2}(x-2)$. Normal line: $y - 4\ln 2 = 2(x-2)$.

25 c)

Tangent: when $y = 0$, $x = 2 + 8\ln 2$. Normal: when $y = 0$, $x = 2 - 2\ln 2$.

25 d)

Height of triangle: $4\ln 2$. Base of triangle: $2 + 8\ln 2 - (2 - 2\ln 2) = 10\ln 2$. Area of triangle: $\frac{1}{2} \times 4\ln 2 \times 10\ln 2 = 20(\ln 2)^2$.

26)

$2x - 2x\frac{dy}{dx} - 2y + 10y\frac{dy}{dx} = 0$. $\frac{dy}{dx} = \frac{2y-2x}{2x-10y} = \frac{y-x}{x-5y}$. Pts A, C: $2x - 2y = 0 \Rightarrow y = x$. $x^2 - 2x(x) + 5(x)^2 = 20 \Rightarrow 4x^2 - 20 = 0$. $\Rightarrow x = \pm\sqrt{5}$ so $C(-\sqrt{5}, -\sqrt{5})$ and $A(\sqrt{5}, \sqrt{5})$. Pts B, D: $x - 5y = 0 \Rightarrow y = \frac{1}{5}x$. $x^2 - 2x(\frac{1}{5}x) + 5(\frac{1}{5}x)^2 = 20 \Rightarrow \frac{4}{5}x^2 - 20 = 0$. $\Rightarrow x = \pm 5$ so $D(-5, -1)$ and $B(5, 1)$.

27 a)

$y = 0 \Rightarrow 0 + 0 = k$ but $k \neq 0$ so it doesn't cross the x-axis.

27 b)

$$2xy\frac{dy}{dx} + y^2 + x^3\frac{dy}{dx} + 3x^2y = 0$$
$$\Rightarrow (2xy + x^3)\frac{dy}{dx} = -y^2 - 3x^2y$$
$$\Rightarrow \frac{dy}{dx} = \frac{-y^2 - 3x^2y}{2xy + x^3} \text{ as req}$$

27 c)

$\left.\frac{dy}{dx}\right|_{x=-1} = \frac{-y^2-3y}{-2y-1}$ stat pt so $-y^2-3y=0$. $\Rightarrow -y(y+3)=0 \Rightarrow y=0$ or $y=-3$. So stat pt at $(-1,0)$ and $(-1,-3)$. But the curve does not cross the x-axis. So $-1\times(-3)^2+(-1)^3(-3)=k$ so $k=-6$.

28 a)

$3+x\frac{dy}{dx}+y+2y\frac{dy}{dx}=0 \Rightarrow \frac{dy}{dx}=-\frac{3+y}{x+2y}$. Parallel to y-axis: $x+2y=0$, therefore $x=-2y$. Therefore $a=-2b$. At (a,b): $3a+ab+b^2=k$.

$$\begin{aligned}&\Rightarrow 3(-2b)+(-2b)b+b^2=k\\&\Rightarrow -6b-2b^2+b^2=k\\&\Rightarrow -6b-b^2=k\end{aligned}$$

At $(5a,5b)$: $15a+25ab+25b^2=k$.

$$\begin{aligned}&\Rightarrow 15(-2b)+25(-2b)b+25b^2=-6b-b^2\\&\Rightarrow -30b-50b^2+25b^2=-6b-b^2\\&\Rightarrow 24b+24b^2=0\\&\Rightarrow 24b(1+b)=0\text{: } b=0 \text{ or } b=-1\end{aligned}$$

Therefore $a=0$ or $a=2$. $(0,0)$ or $(2,-1)$. Find the value of k: $(0,0)$ gives $k=0$ so it does not lie on the curve. $(2,-1)$ gives $6-2+1=k$, therefore $k=5$. Therefore $a=2$, $b=-1$, $k=5$.

28 b)

$(2,-1)$ and $(10,-5)$.

29 a)

LHS: $(-2+3)^4=1$. RHS: $2x-y+8=-4-3+8=1$. So $(-2,3)$ lies on the curve.

29 b)

$4(x+y)^3(1+\frac{dy}{dx})=2-\frac{dy}{dx}$. At $(-2,3)$: $4(-2+3)^3(1+\frac{dy}{dx})=2-\frac{dy}{dx}$.

$$\begin{aligned}&\Rightarrow 4+4\frac{dy}{dx}=2-\frac{dy}{dx}\Rightarrow 5\frac{dy}{dx}=-2\Rightarrow \frac{dy}{dx}=-\tfrac{2}{5}\\&y-3=-\tfrac{2}{5}(x+2)\Rightarrow 5y-15=-2x-4\\&\Rightarrow 2x+5y=11\end{aligned}$$

29 c)

Sub in $x=\frac{11}{2}-\frac{5}{2}y$.

$$\begin{aligned}(\tfrac{11}{2}-\tfrac{5}{2}y+y)^4&=2(\tfrac{11}{2}-\tfrac{5}{2}y)-y+8\\\Rightarrow(\tfrac{11}{2}-\tfrac{3}{2}y)^4&=-6y+19\\\Rightarrow(\tfrac{11}{2}-\tfrac{3}{2}y)^4&=-6y+19\\\Rightarrow \tfrac{14641}{16}-\tfrac{3993}{4}y+\tfrac{3267}{8}y^2-\tfrac{297}{4}y^3+\tfrac{81}{16}y^4&=-6y+19\\\Rightarrow \tfrac{81}{16}y^4-\tfrac{297}{4}y^3+\tfrac{3267}{8}y^2-\tfrac{3969}{4}y+\tfrac{14337}{16}&=0\\\Rightarrow 81y^4-1188y^3+6534y^2-15876y+14337&=0\end{aligned}$$

$$\Rightarrow 27y^4 - 396y^3 + 2178y^2 - 5292y + 4779 = 0$$
$$\Rightarrow (y-3)(27y^3 - 315y^2 + 1233y - 1593) = 0$$
$$\Rightarrow (y-3)(3y^3 - 35y^2 + 137y - 177) = 0$$
$$\Rightarrow (y-3)(y-3)(3y^2 - 26y + 59) = 0$$

$b^2 - 4ac = 676 - 4 \times 3 \times 59 < 0$. Therefore only one real solution of $y = 3$. Therefore the tangent line does not intersect the equation again.

30 a)

$y = \arctan x$ and $\tan y = x$. So $\tan^2 y + 1 = x^2 + 1$. $\sec^2 y \frac{dy}{dx} = 1 \Rightarrow \frac{dy}{dx} = \frac{1}{\sec^2 y} = \frac{1}{1+x^2}$ as req.

30 b)

$\frac{dP}{dt} = \frac{1}{(t-31)^2+1}$.

30 c)

$8.5 + \arctan(t-31) = 9.95$. Let $f(t) = -1.45 + \arctan(t-31)$. $f'(t) = \frac{1}{(t-31)^2+1}$.

$$t_{n+1} = t_n - \frac{-1.45 + \arctan(t_n - 31)}{\frac{1}{(t_n-31)^2+1}}$$

$$t_1 = 30$$
$$t_2 = 34.47079633\ldots$$
$$t_3 = 36.55463056\ldots$$
$$t_4 = 38.38067712\ldots$$
$$t_5 = 39.15024803\ldots$$
$$t_6 = 39.23716974\ldots$$
$$t_7 = 39.23809265\ldots$$

During the year 2004.